

Retail Sales Performance Analysis

Using Excel, SQL Server / T-SQL and Power BI

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Portfolio Project 2 | Business analytics | Currency: USD (\$)

Area	Details
Project type	Retail / business sales analytics case study
Dataset	Synthetic retail sales dataset generated for public portfolio use
Currency	USD (\$)
Tools demonstrated	Excel, SQL Server / T-SQL, Power BI, DAX, dashboard design and business interpretation
Date range analysed	2023-01-01 to 2025-12-30
Records analysed	1,000
Portfolio role alignment	Junior Data Analyst / Business Analyst / Sales Analyst / BI Analyst

1. Executive Summary

This project analyses a synthetic retail sales dataset in order to evaluate sales performance, profit contribution, product-category performance, customer segments, sales channels, returns and monthly performance trends. The project was created as a non-health sector portfolio project to demonstrate that the same analytical skills used in health analytics can also be applied to business, retail, commercial and operations data.

The project follows a practical end-to-end data analyst workflow: preparing the dataset in Excel, validating and querying the data using SQL Server / T-SQL, creating DAX measures in Power BI, and building an interactive dashboard for decision-making. All monetary values in this USD version are expressed in United States Dollars (\$).

2. Problem Statement

Retail managers need clear visibility of revenue, profit, order trends, product performance, customer segments and returns. Without a structured dashboard, it can be difficult to identify which products, regions or channels are driving performance and where corrective action may be needed.

This project addresses the problem by converting raw transactional retail data into a decision-support dashboard that allows users to analyse performance by month, category, customer segment, region and sales channel.

3. Project Objectives

- Analyse sales and profit performance over time.
- Identify top-performing product categories, provinces, channels and segments.
- Calculate key commercial indicators such as net sales, profit margin, average order value and return rate.
- Build an interactive Power BI dashboard using slicers and dynamic measures.
- Prepare a portfolio-ready business analytics case study that is suitable for roles outside the health sector.

4. Key Questions Answered

1. What are the total net sales, profit, order count, average order value and return rate?
2. Which product categories generate the highest sales and profit?
3. Which channels and customer segments contribute the most revenue?
4. How do sales and profit trend over time?
5. Which provinces or regions perform best?
6. Which areas may require action because of low margins or higher returns?

5. Tools Used

Tool	Use in project
Microsoft Excel	Initial review, workbook preparation, summary sheets and dashboard-style output.
SQL Server / T-SQL	Structured querying, aggregation, validation and business-analysis logic.
Power BI	Dashboard development, data modelling, DAX measures, slicers and visual storytelling.
DAX	Measures for net sales, profit, profit margin, AOV, return rate and target achievement.
Python	Synthetic USD sales dataset generation for safe public portfolio use.

6. Dataset Summary

Metric	Value
Currency	USD (\$)
Date range analysed	2023-01-01 to 2025-12-30
Rows analysed	1,000
Customers	500
Products	35
Total net sales	\$185,312.45
Total profit	\$54,477.96
Average order value	\$185.31
Profit margin	29.40%
Return rate	7.30%

7. Data Model

The Power BI model uses a simple star-schema style layout.

Table	Purpose
Fact_Sales	Main transaction table containing order, product, customer, sales, cost, profit, discount and return fields.
Dim_Product	Product lookup table with category, subcategory, base unit price, unit cost and supplier.
Dim_Customer	Customer lookup table with segment, province, region group and city.
Dim_Date	Date table used for year, month, quarter and time-series filtering.
Sales_Targets	Monthly category and region target values for target-achievement analysis.
Returns	Returned order records, reasons and refund amounts.

8. Methodology

Stage	Work performed	Output
1. Data	Generated a synthetic retail sales dataset in USD using	Safe public portfolio dataset.

generation	Python.	
2. Excel preparation	Organised fact, dimension and summary sheets.	Excel workbook.
3. SQL analysis	Created SQL Server / T-SQL queries for sales, profit, category, channel and trend analysis.	SQL script.
4. Power BI modelling	Loaded fact and dimension tables and created relationships.	Power BI data model.
5. DAX measures	Created KPI measures for revenue, profit, margin, AOV, returns and targets.	Reusable DAX measures.
6. Dashboard design	Built pages for executive overview, sales trends, product/category analysis and customer/region insights.	Interactive Power BI dashboard.
7. Interpretation	Converted analysis outputs into findings and recommendations.	Case study and README.

9. Core Calculations and DAX Measures

Measure	Purpose	DAX / logic
Net Sales	Revenue after discount	SUM(Fact_Sales[Net_Sales_USD])
Gross Sales	Revenue before discount	SUM(Fact_Sales[Gross_Sales_USD])
Total Profit	Net sales less total cost	SUM(Fact_Sales[Profit_USD])
Profit Margin	Profit as a percentage of net sales	DIVIDE([Total Profit], [Net Sales], 0)
Orders	Unique order count	DISTINCTCOUNT(Fact_Sales[Order_ID])
Average Order Value	Average revenue per order	DIVIDE([Net Sales], [Orders], 0)
Return Rate	Returned orders as a percentage of total orders	DIVIDE([Returned Orders], [Orders], 0)

10. Power BI Dashboard Pages

Page	Purpose	Main visuals
Executive Overview	High-level sales and profitability view.	KPI cards, monthly sales trend, category and channel charts.
Sales Trends	Shows monthly sales, profit and order movement.	Line/column charts and year/month slicers.
Product & Category Analysis	Identifies strongest categories, subcategories and products.	Top products, category contribution and profit margin visuals.
Customer & Region Insights	Compares customer segments, provinces and channels.	Segment charts, map/province chart and customer table.
Insights & Recommendations	Summarises findings, risks and recommended actions.	Text boxes and business interpretation.

11. Selected Outputs

The dashboard uses USD-based sales and profitability measures. Example KPI outputs from the generated dataset include:

KPI	Value
Total Net Sales	\$185,312.45
Total Profit	\$54,477.96
Orders	1,000
Average Order Value	\$185.31
Profit Margin	29.40%

Return Rate	7.30%
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12. Key Findings

- The dataset produced total net sales of \$185,312.45 and total profit of \$54,477.96.
- Technology was the highest net-sales category in the generated dataset.
- Online was the strongest sales channel by net sales.
- Profit margin varies by product category, showing that high sales volume does not always mean the highest profitability.
- Returns affect revenue quality and should be monitored together with sales and profit.
- Monthly trends show seasonal changes, which supports planning for stock, promotions and sales targets.

13. Business Interpretation

- Managers should monitor both sales and profit because a high-selling product may have lower margins.
- Channel analysis helps identify where marketing and sales effort is producing the best returns.
- Segment analysis supports targeted sales strategies for consumer, corporate and small-business customers.
- Return-rate tracking helps identify possible quality, delivery or customer-expectation issues.
- The dashboard can help business users move from static spreadsheet review to interactive decision-making.

14. Data Limitations

- The dataset is synthetic and was generated for portfolio use; it does not represent a real company.
- The analysis is designed to demonstrate skills, not to make claims about an actual market.
- Sales performance is affected by assumed product mix, pricing, discounts and seasonal patterns.
- Future extensions could include real public datasets, advertising spend, inventory, customer acquisition cost and geographic mapping.

15. Recommendations

- Track net sales, profit margin and return rate together instead of focusing only on revenue.
- Investigate product categories with strong sales but weaker margins.
- Use channel performance to guide sales and marketing investment.
- Monitor returns by product and reason to reduce avoidable refunds.
- Extend the dashboard with targets, forecasting and inventory analysis in future portfolio work.

16. Project Outcome

This project demonstrates the ability to work with business data outside the health sector. It shows practical use of Excel, SQL Server / T-SQL and Power BI to analyse sales, profit, customers, products, channels and returns. It strengthens the portfolio by proving that the analyst can apply data skills across industries, not only in healthcare.